

Review of the US Patent Literature on Digital Twins and Possible Applications for Coordinate Metrology

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Abstract

The idea of “digital twins” is relatively new—although it will be argued that the coordinate metrology community has in fact been using the concepts of “digital twins” for many years. The term was first introduced by Michael Grieves, in the context of product lifecycle management (PLM), at a Society of Manufacturing Engineers conference in Troy Michigan in 2002. A review of the literature shows there has historically been little agreement on the definition of digital twins or nomenclature used to describe them—however the Digital Twin Consortium was launched in 2020 by member companies to focus on technology and terminology to drive adoption of digital twins. There does seem to be agreement that, at a minimum, there is a connection between a physical device and a virtual model of the device, and the connection requires real-time data derived from sensors processed by automated software. However, there seems to be little discussion in the literature about the parameters in the model and how to determine the optimum sensor architecture to measure the parameters—and almost nothing about the uncertainty analysis. While digital twins are an active field of academic research, a US patent search shows that practical applications have been slow in developing, with the first two applications being filed in 2015. A review of the US patent literature is presented, with an emphasis on applications that could possibly benefit from expertise of the CMSC community.

Index Terms

digital twin, digital model, digital shadow

I. INTRODUCTION

The idea of digital twins is relatively new. First introduced by Michael Grieves, in the context of product lifecycle management (PLM), at a Society of Manufacturing Engineers conference in Troy Michigan in 2002 [1], [2]. Grieves talked about the evolution of the notion of digital twins in a video interview with Machine Design [3], and an audio interview with Beyond BIM [4], in 2021.

A review of the literature shows that there is little agreement on the definition of a digital twin or nomenclature; but rather each researcher and marketing department has historically defined their own terms [5], [6]. Recognizing the need for an organization to represent digital twins, the Digital Twin Consortium was founded by 19 sponsoring companies in May 2020. According to the list of members on their website, there are now 153 member companies. They have 11 working groups, including: Academia & Research; Aerospace & Defense; Architecture, Engineering, Construction & Operations; and Technology, Terminology & Taxonomy. They have identified 8 initiatives, including: Definition of a Digital Twin, and Glossary of Digital Twins. Dan Isaacs, CTO of the Digital Twin Consortium, participated in a webinar sponsored by an intellectual property law firm [7], in which he gave interesting examples of digital twins, e.g., self-driving cars.

There does seem to be agreement that at a minimum there is a connection between a physical device and a virtual model of the device, and the connection requires real-time data derived from sensors processed by automated software. However, it has been argued by the author [8], [9] that there seems to be little discussion in the literature about the parameters in the model and how to determine the optimum sensor architecture to measure the parameters—and almost nothing about the uncertainty analysis. For example, a bridge engineer may want the model to include absolute coordinates of cardinal points, differential

movement of the cardinal points, differential movement between cardinal points, vibrational modes, thermal expansion, wind forces, etc. Only after the model parameters are determined can the sensor architecture, required sensitivities, and uncertainties be determined.

Fuller et al. gives six definitions in section II.A. of their paper [10]. Probably more relevant to the coordinate metrology community, they talk about Digital Twin Misconceptions in section II.B. in which they contrast between a digital model, a digital shadow, and a digital twin.

On December 3, 2020, the Digital Twin Consortium released the following definition:

A digital twin is a virtual representation of real-world entities and processes, synchronized at a specified frequency and fidelity.

Note that in the two cited interviews with Grieves, he placed a broader emphasis on any digital information that is of value, rather than a rigid definition.

It is noteworthy that among the present 153 consortium members, no coordinate metrology instrument companies, national metrology laboratories, or coordinate service providers are represented—even though it will be argued that coordinate metrologist could make significant contributions in at least the four cited working groups.

II. DIGITAL TWIN SEARCH

A US patent search was conducted on June 6, 2022 for all patents and published patent applications that contained the words “digital” and “twin” within 3 words of each other, in the abstracts. Note that applications are published 18 months from the application date, unless early publication is requested, or it claims priority to an earlier application. So, the search on June 6 would include at least all applications filed before December 6, 2020.

A total of 271 patents and applications met the search criteria, with the following breakdown of numbers by application year: 2015-2, 2016-6, 2017-15, 2018-35, 2019-54, 2020-84 (partial year), and 2021-74 (partial year). The major assignees include: GE-27, IBM-30, Siemens-23, Strong Force-28, and Toyota-11.

The titles of all 271 patents/applications were examined. The abstracts of titles that indicated a possible connection with coordinate metrology were read. The full specifications were printed and examined for those that the abstracts indicated a possible connection with coordinate metrology. After many hours of examination, and approximately two reams of printer paper, one patent, to Garvey et al. [11] at Grale Technologies, and two pending applications—one to Garvey et al. [12] for a continuation of the patent, and one to Zhang et al. [13] at GE Inspection Technologies, LP, were connected to coordinate metrology, i.e., 3/271. The Grale Technologies patent and continuing application are drawn to measurements made while the finished product is still in the work cell, e.g., while still in the lathe, milling machine, etc. The GE Inspection Technologies application is drawn to ultrasonic testing.

While it may appear, based on the small number of US patents, that digital twins have little application to coordinate metrology, it will be argued that in fact, coordinate metrologist have been employing the principles of digital twins for many years—they just haven’t adopted the popular nomenclature. The historical absence of a similar term makes a prior art search very difficult. For example, the patent examiner for the Grale Technologies patent [11] did not cite any prior art (as indicated by the absence of an * beside the patent numbers on the front page of the patent) and it was allowed in a first office action.

III. EXAMPLES OF US PATENTS THAT DISCLOSE COORDINATE METROLOGY DIGITAL TWINS

It will be argued that in fact there are many examples of US patents that disclose applications of coordinate metrology and what now could be called digital twins. Based on previous knowledge of the art, rather than a patent search, a representative list of 37 patents is included in Appendix A. Note that this is a fraction, and there are probably hundreds, if not thousands, of additional examples in the US patent literature.

It is interesting that many of the general purpose coordinate metrology instruments in the CMSC exhibit hall originated by inventors solving specific coordinate metrology problems, i.e., necessity is the mother of invention. For example, the articulated arm coordinate measurement machines (AACMM) were pioneered by inventors including Eaton [14], [15] in order to create a digital twin of bent tubing; Fraser & Raab [16]–[20] to create a digital twin of a skeletal system and other orthopedic applications, and later for such things as a digital twin of surfaces, tube bending, and robots [21]–[23]; and others.

The laser tracker was pioneered by inventors including Greenleaf [24] to create a digital twin of surfaces; Merry, Brown & Wells [25], [26], Lau & Hocken [27], and Greenwood & Pastusak [28]–[30] to create digital twins of precision machines; Payne, Parker & Bradley [31] to create a digital twin of a radio telescope; and others.

The laser scanner was pioneered by inventors including Crampton [32]; Riegl, Ullrich & Zierlinger [33]; Gittinger, Becker, & Becker [34] to create digital twins of surfaces; and others.

Lesser known examples include Parker, Caruso, Thrasher & Blinn [35], [36] for creating a digital twin of tire noise based on the tread geometry; Parker & Payne [37]–[41] for creating a digital twin of a laser tracker, civil structures, pressure vessels, and concrete; Marsh & Vanscotter [42], [43] for creating a digital twin of an aircraft structure; Calkins & Salerno [44] for creating a digital twin of a robot; Grau & Parker [45] for creating a digital twin of a part; Fuchs [46] for creating a digital twin of a bridge girder and bolted joints; Pettersson [47] for creating a digital twin of building parts; Wu, Yi, Zhao, Zhang, Xu & Zhu [48]–[51] for creating digital twins for structural health monitoring; and others.

IV. CONCLUSIONS

The coordinate metrology community is actually leading in practical applications of digital twins, but there is a communications gap with other communities. The Coordinate Metrology Society may need to collaborate with the Digital Twin Consortium, to make them more aware of the state-of-the-art of dimensional metrology. Instrument companies and coordinate service providers may want to adopt the “digital twin” nomenclature in order to expand their markets.

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APPENDIX A

	A	C	D	F	G	I
	Document Numb	Publication Date	Title	Inventor Name	Assignee	Filing Date
1	3944798	1976-03-16	Method and apparatus for measuring a laser beam	Eaton, Homer L. (Balboa, CA)	Eaton-Leonard Corporation (San Francisco, CA)	1974-04-18
2	4457625	1984-07-03	Self calibrating contour measuring system	Greenleaf, Allen H. (Lexington, MA)	Itek Corporation (Lexington, MA)	1981-07-13
3	4621926	1986-11-11	Interferometer system for controlling a laser beam	Merry, Bradford J. (Annapolis, MD)	Lasercon Corporation (Lanham, MD)	1985-04-30
4	4649934	1987-03-17	Joint laxity measurement	Fraser, Gregory A. (Dollard Des Ormes, France)	Faro Medical Technologies, Inc. (Farmingdale, NY)	1985-06-07
5	4714339	1987-12-22	Three and five axis laser tracking system	Lau, Kam C. (Gaithersburg, MD)	The United States of America (Washington, DC)	1986-02-28
6	4727501	1988-02-23	Method of simulating tire tread	Parker, David H. (Brecksville, OH)	The Uniroyal Goodrich Tire Company (Akron, OH)	1987-02-27
7	4760851	1988-08-02	3-dimensional digitizer for skeletal analysis	Fraser, Gregory A. (Dollard des Ormes, France)	Faro Medical Technologies Inc. (Farmingdale, NY)	1986-03-31
8	4788651	1988-11-29	Method for improving tread noise	Parker, David H. (Brecksville, OH)	The Uniroyal Goodrich Tire Company (Akron, OH)	1988-02-22
9	4790651	1988-12-13	Tracking laser interferometer	Brown, Lawrence B. (Annapolis, MD)	Chesapeake Laser Systems, Inc. (Annapolis, MD)	1987-09-30
10	5305203	1994-04-19	Computer-aided surgery apparatus	Raab, Simon (Lorraine, CA)	Faro Medical Technologies Inc. (Farmingdale, NY)	1990-10-02
11	5408754	1995-04-25	Method and apparatus for measuring a laser beam	Raab, Simon (Longwood, FL)	Faro Technologies, Inc. (Lake Mary, FL)	1993-10-26
12	5412880	1995-05-09	Method of constructing a 3-dimensional model	Raab, Simon (Longwood, FL)	Faro Technologies Inc. (Lake Mary, FL)	1993-08-26
13	5455670	1995-10-03	Optical electronic distance measuring system	Payne, John M. (Tucson, AZ)	FAssociated Universities, Inc. (Washington, DC)	1993-05-27
14	5748767	1998-05-05	Computer-aided surgery apparatus	Raab, Simon (Longwood, FL)	Faro Technology, Inc. (Lake Mary, FL)	1993-08-10
15	5768792	1998-06-23	Method and apparatus for measuring a laser beam	Raab, Simon (Longwood, FL)	Faro Technologies Inc. (Lake Mary, FL)	1996-02-09
16	5903459	1999-05-11	Method for product acceptance	Greenwood, Thomas A. (Sumner, WA)	The Boeing Company (Seattle, WA)	1997-06-03
17	5920483	1999-07-06	Scaling machine media for thermal analysis	Greenwood, Thomas A. (Sumner, WA)	The Boeing Company (Seattle, WA)	1997-06-03
18	5949685	1999-09-07	Real-time orientation of machine tool	Greenwood, Thomas A. (Sumner, WA)	The Boeing Company (Seattle, WA)	1997-06-03
19	6535794	2003-03-18	Method of generating an error map	Raab, Simon (Longwood, FL)	Faro Technologies Inc. (Lake Mary, FL)	1995-05-03
20	6611617	2003-08-26	Scanning apparatus and method	Crampton, Stephen James (Washington, DC)	CRAMPTON STEPHEN JAMES (Washington, DC)	1998-05-26
21	6852975	2005-02-08	Method for the recording of an object	Riegl, Johannes (Trautenberg, Austria)	RIEGL Laser Measurement Systems (Vienna, Austria)	2002-10-04
22	7672817	2010-03-02	Flight in factory	Marsh, Bob (Lake Stevens, WA)	The Boeing Company (Chicago, IL)	2006-11-08
23	7847922	2010-12-07	Method and an apparatus for calibrating a laser beam	Gittinger, Juergen (Ludwigsburg, Germany)	Faro Technologies, Inc. (Lake Mary, FL)	2007-06-29
24	7856334	2010-12-21	Method for calibrating a laser beam	Parker, David H. (Earlysville, VA, US)		2008-12-06
25	7895015	2011-02-22	Method for measuring the structure of a laser beam	Parker, David H. (Earlysville, VA, US)	John M. (Charlottesville, VA, US)	2008-12-04
26	8209134	2012-06-26	Methods for modeling the structure of a laser beam	Parker, David H. (Earlysville, VA, US)	Solliday, Laura P. (Green Bank, WV)	2010-12-20
27	8688408	2014-04-01	Flight in factory	Marsh, Bob J. (Lake Stevens, WA)	The Boeing Company (Chicago, IL)	2009-10-09
28	9354043	2016-05-31	Methods for measuring and modeling a laser beam	Parker, David H. (Earlysville, VA, US)	Solliday, Laura P. (Green Bank, WV)	2012-05-31
29	9452533	2016-09-27	Robot modeling and positioning	Calkins, Joseph M. (Williamsburg, VA)	New River Kinematics, Inc. (Williamsburg, VA)	2013-05-15
30	9638507	2017-05-02	Measurement machine utilizing a laser beam	Grau, Markus (Kornthal-Muenchen, Germany)	FARO Technologies, Inc. (Lake Mary, FL)	2013-01-22
31	9803977	2017-10-31	Apparatus and method for bridging a gap	Fuchs, Paul A. (Leesburg, VA, US)	Fuchs Consulting, Inc. (Leesburg, VA)	2014-01-22
32	9934335	2018-04-03	Method and system for virtual assembly	Pettersson, Bo (London, GB)	HEXAGON TECHNOLOGY CE (London, GB)	2013-10-07
33	9964468	2018-05-08	Optimizing sensor placement for a laser beam	Wu, Zheng Yi (Watertown, CT)	Bentley Systems, Incorporated (Watertown, CT)	2014-12-08
34	9983776	2018-05-29	Software system for dynamic feedback	Wu, Zheng Yi (Watertown, CT)	Bentley Systems, Incorporated (Watertown, CT)	2016-06-30
35	10203268	2019-02-12	Methods for measuring and modeling a laser beam	Parker, David H. (Earlysville, VA, US)	Solliday, Laura P. (Green Bank, WV)	2016-05-26
36	10295435	2019-05-21	Model-based damage detection	Wu, Zheng Y. (Watertown, CT)	Bentley Systems, Incorporated (Watertown, CT)	2015-06-17
37	10451416	2019-10-22	Optimizing sensor placement for a laser beam	Wu, Zheng Yi (Watertown, CT)	Bentley Systems, Incorporated (Watertown, CT)	2016-06-20